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SPHERICAL MIRRORS

A spherical reflecting surface has image-forming properties similar to those of a thin lens or of a single refracting surface. The image from a spherical mirror is in some respects superior to that from a lens, notably in the absence of chromatic effects due to dispersion that always accompany the refraction of white light. Therefore mirrors are occasionally used in place of lenses in optical instruments, but their applications are not so broad as those of lenses because they do not offer the same possibilities for correction of the other aberrations of the image (see Chap. 9).

Because of the simplicity of the law of reflection compared with the law of refraction, the quantitative study of image formation by mirrors is easier than for lenses. Many features are the same, and these we shall pass over rapidly, putting the chief emphasis upon those characteristics which are different. To begin with, we restrict the discussion to images formed by paraxial rays.

6.1 FOCAL POINT AND FOCAL LENGTH

Diagrams showing the reflection of a parallel beam of light by a concave mirror and by a convex one are given in Fig. 6A. A ray striking the mirror at some point such as T obeys the law of reflection $\phi'' = \phi$. All rays are shown as brought to a common

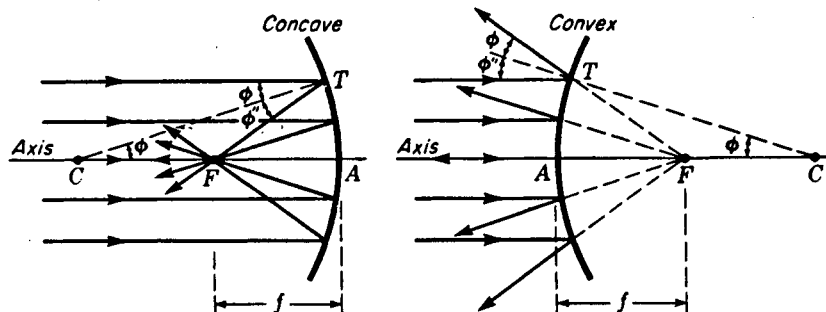


FIGURE 6A

The primary and secondary focal points of spherical mirrors coincide.

focus at F , although this will be strictly true only for paraxial rays. The point F is called the *focal point* and the distance FA the *focal length*. In the second diagram the reflected rays diverge as though they came from a common point F . Since the angle TCA also equals ϕ , the triangle TCF is isosceles, and in general $CF = FT$. But for very small angles ϕ (paraxial rays), FT approaches equality with FA . Hence

$$FA = \frac{1}{2}(CA)$$

$$f = -\frac{1}{2}r \quad (6a)$$

and the focal length equals one-half the radius of curvature [see also Eq. (6d)].

The negative sign is introduced in Eq. (6a) so that the focal length of a concave mirror, which behaves like a positive or converging lens, will also be positive. According to the sign convention of Sec. 3.5, the radius of curvature is negative in this case. The focal length of a convex mirror, which has a positive radius, will then come out to be negative. This sign convention is chosen as being consistent with that used for lenses; it gives converging properties to a mirror with positive f and diverging properties to a mirror with negative f . By the principle of reversibility it can be seen from Fig. 6A that the primary and secondary focal points of a mirror coincide. In other words, it has but one focal point.

As before, a transverse plane through the focal point is called the focal plane. Its properties, as shown in Fig. 6B, are similar to those of either focal plane of a lens; e.g., parallel rays incident at any angle with the optic axis are brought to a focus at some point in the focal plane. The image Q' of a distant off-axis point object occurs at the intersection with the focal plane of that ray which goes through the center of curvature C .

6.2 GRAPHICAL CONSTRUCTIONS

Figure 6C, which illustrates the formation of a real image by a concave mirror, is self-explanatory. When the object MQ is moved toward the center of curvature C , the image also approaches C and increases in size until when it reaches C , it is the

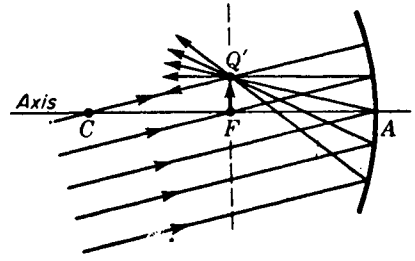


FIGURE 6B
Parallel rays incident on a concave mirror but inclined to the axis are brought to a focus in the focal plane.

same size as the object. The conditions when the object is between C and F can be deduced from the interchangeability of object and image as applied to this diagram. When the object is inside the focal point, the image is virtual, as with a converging lens. The methods of graphically constructing the image follow the same principles as those used for lenses, including the fact that paraxial rays must be represented as being deflected at the tangent plane instead of at the actual surface.

An interesting experiment can be performed with a large concave mirror set up under the condition of unit magnification, as shown in Fig. 6D. A bouquet of flowers is suspended upside down in a box and illuminated by a shaded lamp S . The large mirror is placed with its center of curvature C at the top surface of the stand, on which a real vase is placed. The observer's eye at E sees a perfect reproduction of the bouquet, not merely as a picture but as a faithful three-dimensional replica, which creates a strong illusion that it is a real object. As shown in the diagram, the rays diverge from points on the image just as they would if the real object were in the same position.

The parallel-ray method of construction is given for a concave mirror in Fig. 6E. Three rays leaving Q are, after reflection, brought to the conjugate point Q' . The image is real, inverted, and smaller than the object. Ray 4 drawn parallel to the axis is, by definition of the focal point, reflected through F . Ray 6 drawn through F is reflected parallel to the axis, and ray 8 through the center of curvature strikes the

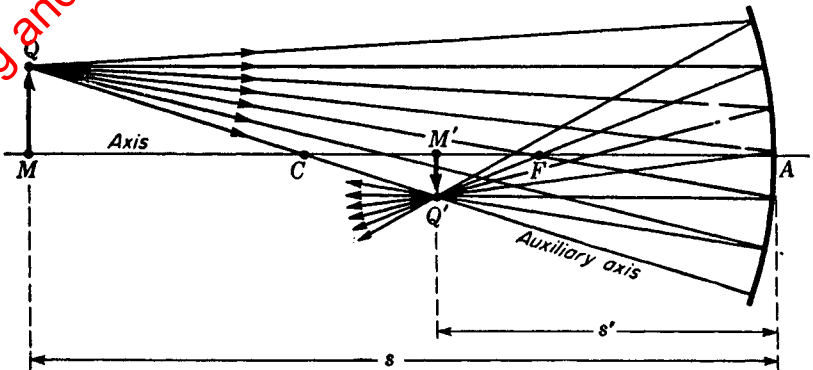


FIGURE 6C
Real image due to a concave mirror.

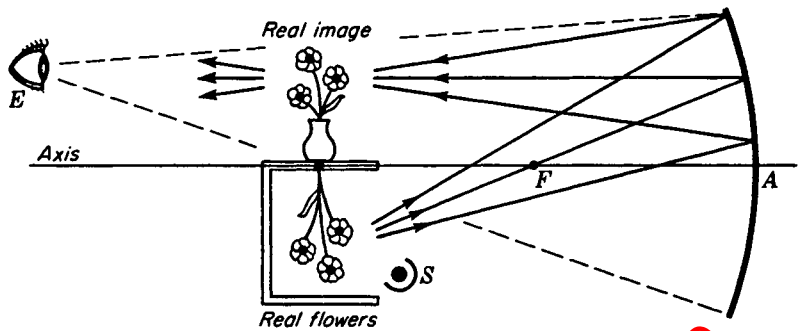


FIGURE 6D

Experimental arrangement for an optical illusion produced by a real image of unit magnification. The three-dimensional image shows parallax, as the real flowers do, and the image is so real that the eye cannot detect the difference between the real image and the real object.

mirror normally and is reflected back on itself. The crossing point of any two of these rays is sufficient to locate the image.

A similar procedure is applied to a convex mirror in Fig. 6F. The rays from the object point Q , after reflection, diverge from the conjugate point Q' . Ray 4, starting parallel to the axis, is reflected as if it came from F . Ray 6 toward the center of curvature C is reflected back on itself, while ray 7 going toward F is reflected parallel to the axis. Since the rays never pass through Q' , the image $Q'M'$ in this case is virtual.

The oblique-ray method can also be used for mirrors, as illustrated in Fig. 6G for a concave mirror. After drawing the axis 1 and the mirror 2, we lay out the points C and F and draw a ray 3 making any arbitrary angle with the axis. Through F , the broken line 4 is then drawn parallel to 3. Where this line intersects the mirror at S , a parallel ray 6 is drawn backward to intersect the focal plane at P . Ray 7 is then drawn through TP and intersects the axis at M' . By this construction M and M' are conjugate points and 3 and 7 are the parts of the ray in object and image spaces. The principle involved in this construction is obvious from the fact that if 3 and 4

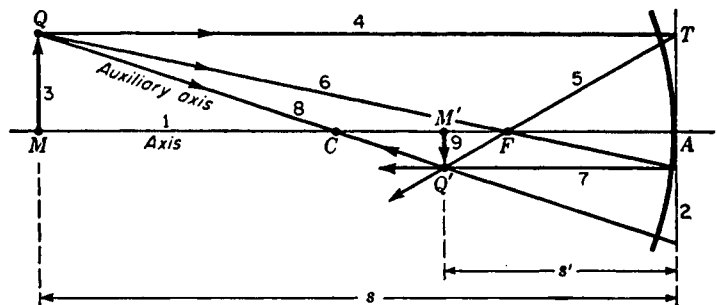


FIGURE 6E

Parallel-ray method for graphically locating the image formed by a concave mirror.

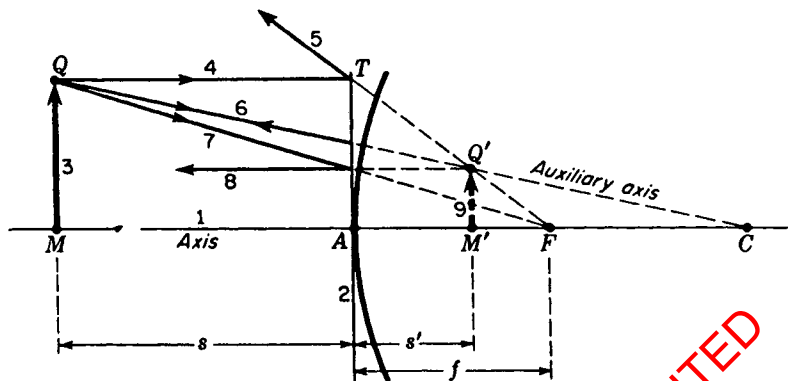


FIGURE 6F
Parallel-ray method for graphically locating the image formed by a convex mirror.

were parallel incident rays, they would come to a focus at F in the focal plane. If in place of ray 4 another ray were drawn through C and parallel to ray 3, it too would cross the focal plane at P . A ray through the center of curvature would be reflected directly back upon itself.

6.3 MIRROR FORMULAS

In order to be able to apply the standard lens formulas of the preceding chapters to spherical mirrors with as little change as possible, we must adhere to the following sign conventions:

- 1 Distances measured from left to right are positive while those measured from right to left are negative.

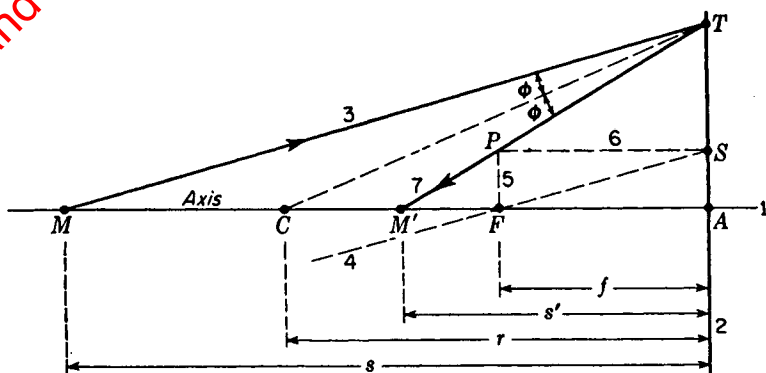


FIGURE 6G
Oblique-ray method for locating the image formed by a concave mirror.

- 2 Incident rays travel from left to right and reflected rays from right to left.
- 3 The focal length is measured from the focal point to the vertex. This gives f a positive sign for concave mirrors and a negative sign for convex mirrors.
- 4 The radius is measured from the vertex to the center of curvature. This makes r negative for concave mirrors and positive for convex mirrors.
- 5 Object distances s and image distances s' are measured from the object and from the image respectively to the vertex. This makes both s and s' positive and the object and image real when they lie to the left of the vertex; they are negative and virtual when they lie to the right.

The last of these sign conventions implies that for mirrors the object space and the image space coincide completely, the actual rays of light always lying in the space to the left of the mirror. Since the refractive index of the image space is the same as that of the object space, the n' of the previous equations becomes numerically equal to n .

The following is a simple derivation of the formula giving the conjugate relations for a mirror. In Fig. 6G it is observed that by the law of reflection the radius CT bisects the angle MTM' . Using a well-known geometrical theorem, we can then write the proportion

$$\frac{MC}{MT} = \frac{CM'}{M'T}$$

Now, for paraxial rays, $MT \approx MA = r$ and $M'T \approx M'A = s'$, where the symbol $\approx$ means "is approximately equal to." Also, from the diagram,

$$MC = MA - CA = s + r$$

and

$$CM' = CA - M'A = -r - s' = -(s' + r)$$

Substituting in the above proportion gives

$$\frac{s + r}{s} = -\frac{s' + r}{s'}$$

which can easily be put in the form

$$\frac{1}{s} + \frac{1}{s'} = -\frac{2}{r} \quad (6b)$$

Mirror formula

The primary focal point is defined as that axial object point for which the image is formed at infinity, so substituting $s = f$ and $s' = \infty$ in Eq. (6b), we have

$$\frac{1}{f} + \frac{1}{\infty} = -\frac{2}{r}$$

from which

$$\frac{1}{f} = -\frac{2}{r} \quad \text{or} \quad f = -\frac{r}{2} \quad (6c)$$

The secondary focal point is defined as the image point of an infinitely distant object point. This is $s' = f'$ and $s = \infty$, so that

$$\frac{1}{\infty} + \frac{1}{f'} = -\frac{2}{r}$$

from which
$$\frac{1}{f'} = -\frac{2}{r} \quad \text{or} \quad f' = -\frac{r}{2} \quad (6d)$$

Therefore the primary and secondary focal points fall together, and the magnitude of the focal length is one-half the radius of curvature. When $-2/r$ is replaced by $1/f$, Eq. (6b) becomes

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \quad (6e)$$

just as for lenses.

The lateral magnification of the image from a mirror can be evaluated from the geometry of Fig. 6C. From the proportionality of sides in the similar triangles $Q'AM'$ and QAM we find that $-y'/y = s'/s$, giving

$$m = \frac{y'}{y} = -\frac{s'}{s} \quad (6f)$$

EXAMPLE 1 An object 2.0 cm high is situated 10.0 cm in front of a concave mirror of radius 16.0 cm. Find (a) the focal length of the mirror, (b) the position of the image, and (c) the lateral magnification.

SOLUTION The given quantities are $y = +2.0$ cm, $s = +10.0$ cm, and $r = -16.0$ cm. The unknown quantities are f , s' , and m . (a) By Eq. (6c),

$$f = -\frac{-16}{2} = +8.0 \text{ cm}$$

(b) By Eq. (6e),

$$\frac{1}{10} + \frac{1}{s'} = \frac{1}{8} \quad \text{or} \quad \frac{1}{s'} = \frac{1}{8} - \frac{1}{10} = \frac{1}{40}$$

giving

$$s' = +40.0 \text{ cm}$$

(c) By Eq. (6f),

$$m = -\frac{40}{10} = -4$$

The image occurs 40.0 cm to the left of the mirror, is 4 times the size of the object, and is real and inverted.

6.4 POWER OF MIRRORS

The power notation that was used in Sec. 4.12 to describe the image-forming properties of lenses can readily be extended to spherical mirrors as follows. As definitions, we let

$$P = \frac{1}{f} \quad V = \frac{1}{s} \quad V' = \frac{1}{s'} \quad K = \frac{1}{r} \quad (6g)$$

Equations (6b), (6e), (6c), and (6f) then take the forms

$$V + V' = -2K \quad (6h)$$

$$V + V' = P \quad (6i)$$

$$P = -2K \quad (6j)$$

$$m = \frac{y'}{y} = -\frac{V}{V'} \quad (6k)$$

EXAMPLE 2 An object is located 20.0 cm in front of a convex mirror of radius 50.0 cm. Calculate (a) the power of the mirror, (b) the position of the image, and (c) its magnification.

SOLUTION Expressing all distances in meters, we have

$$K = \frac{1}{0.50} = +2 \text{ D} \quad \text{and} \quad V = \frac{1}{0.20} = +5 \text{ D}$$

(a) By Eq. (6j),

$$P = -2K = -4 \text{ D}$$

(b) By Eq. (6i),

$$5 + V' = -4 \quad \text{or} \quad V' = -9 \text{ D}$$

or

$$s' = \frac{1}{V'} = \frac{1}{-9} = -0.111 \text{ m} = -11.1 \text{ cm}$$

(c) By Eq. (6k),

$$m = -\frac{5}{-9} = +0.555$$

The power $P = -4 \text{ D}$, and the image is virtual and erect. It is located 11.1 cm to the right of the mirror and has a magnification of $0.555\times$.

6.5 THICK MIRRORS

The term *thick mirror* is applied to a lens system in which one of the spherical surfaces is a reflector. Under these circumstances the light passing through the system is reflected by the mirror back through the lens system, from which it emerges finally into the space from which it entered the lens. Three common forms of optical systems that may be classified as thick mirrors are shown in Fig. 6H. In each case the surface farthest to the right has been drawn with a heavier line than the others, designating the reflecting surface. A parallel incident ray is also traced through each system to where it crosses the axis, thus locating the focal point.

In addition to a focal point and focal plane every thick mirror has a principal point and a principal plane. Two graphical methods by which principal points and planes can be located are given below. The *oblique-ray method* is applied to (a) the thin lens and mirror combination in Fig. 6I, while the *auxiliary-diagram method* is applied to (b) the thick lens and mirror combination in Fig. 6J.

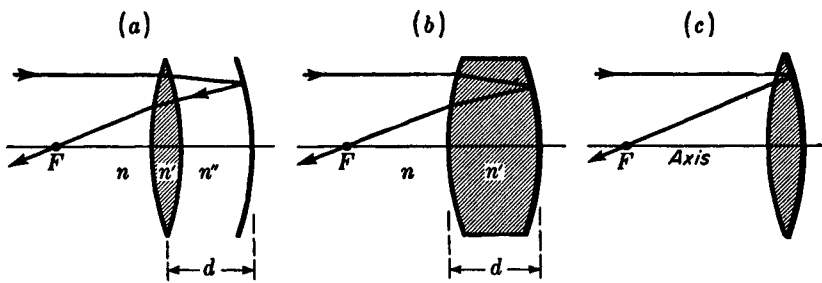


FIGURE 6H
Diagrams of several types of thick mirrors, showing the location of their respective focal points.

In the first illustration the lens is considered *thin* so that its own principal points may be assumed to coincide at H_1 , its center. An incident ray parallel to the axis is refracted by the lens, reflected by the mirror, and again refracted by the lens before it crosses the axis of the system at F . The point T where the incident and final rays, when extended, cross each other locates the principal plane, and H represents the principal point. If we follow the sign conventions for a single mirror (Sec. 6.3), the focal length f of this particular combination is positive and is given by the interval FH .

In the second illustration (Fig. 6J) the incident ray is refracted by the first surface, reflected by the second, and finally refracted a second time by the first surface to a point F where it crosses the axis. The point T where the incident and final rays intersect locates the principal plane and principal point H .

The graphical ray-tracing construction for this case, shown in the auxiliary diagram in Fig. 6J, is started by drawing XZ parallel to the axis. With the origin O near the center, intervals proportional to n and n' are measured off in both directions

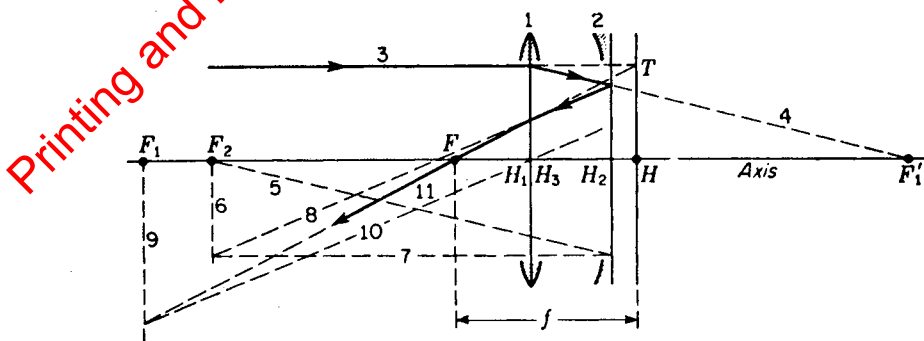


FIGURE 6J
Oblique-ray construction for locating the focal point and principal point of a thick mirror.

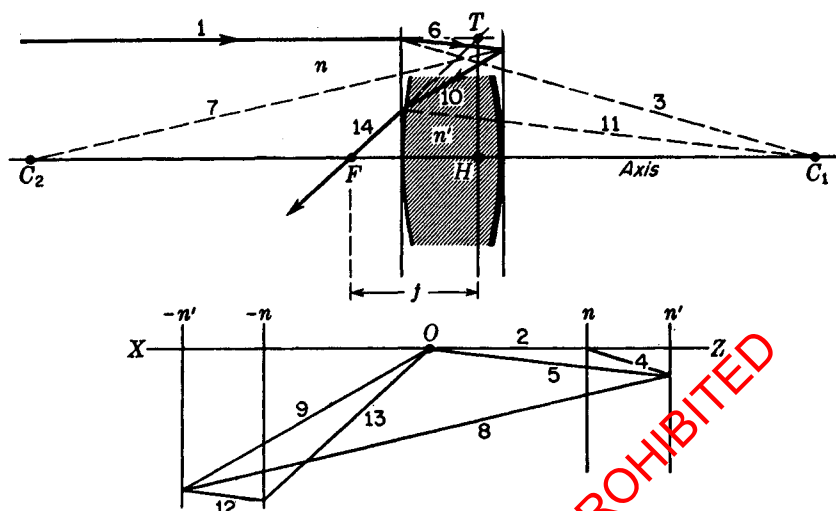


FIGURE 6J
Auxiliary-diagram method of graphically locating the focal point and principal point of a thick mirror.

along XZ . After the vertical lines representing n and n' are drawn, the remaining lines are drawn in the order of the numbers 1, 2, 3, Each even-numbered line is drawn parallel to its preceding odd-numbered line. The proof that this construction is exact for paraxial rays is similar to that given for Fig. 3I.

6.6 THICK-MIRROR FORMULAS

These formulas will be given in the power notation for case (a) shown in Fig. 6H. When r_1 , r_2 , and r_3 are the radii of the three surfaces consecutively from left to right, the power of the combination can be shown* to be given by

$$\bullet \quad P = (1 - cP_1)(2P_1 + P_2 - cP_1P_2) \quad (6l)$$

where, for the case in diagram (a) only and $n'' = n$,

$$P_1 = (n' - n)(K_1 - K_2) \quad (6m)$$

$$P_2 = -2nK_3 \quad (6n)$$

and

$$K_1 = \frac{1}{r_1} \quad K_2 = \frac{1}{r_2} \quad K_3 = \frac{1}{r_3}$$

* For a derivation of these equations, see J. P. C. Southall, "Mirrors, Prisms, and Lenses," 3d ed., p. 379, The Macmillan Company, New York, 1936.

[see Eqs. (4p) and (6d)]. Of the refractive indices, n' represents that of the lens and n that of the surrounding space. The distance from the lens to the principal point of the combination is given by

$$H_1H = \frac{c}{1 - cP_1} \quad (6o)$$

where H_1 is located at the center of the lens and $c = \frac{d}{n}$ (6p)

It is important to note from Eq. (6o) that the position of H is independent of the power P_2 of the mirror and therefore of its curvature K_3 .

EXAMPLE 3 A thick mirror like that shown in Fig. 6H(a) has as one component a thin lens of index $n' = 1.50$, radii $r_1 = +50.0$ cm, and $r_2 = -50.0$ cm. This lens is situated 10.0 cm in front of a mirror of radius -50.0 cm. Assuming that air surrounds both components, find (a) the power of the combination, (b) the focal length, and (c) the principal point.

SOLUTION (a) By Eq. (6m), the power of the lens is

$$P_1 = (1.50 - 1) \left(\frac{1}{0.50} - \frac{1}{-0.50} \right) = +2 \text{ D}$$

Equation (6n) gives for the power of the mirror

$$P_2 = -2 \frac{1}{-0.50} = +4 \text{ D}$$

From Eq. (6p),

$$c = \frac{d}{n} = \frac{0.10}{1} = 0.10 \text{ m}$$

Finally the power of the combination is given by Eq. (6l) as

$$\begin{aligned} P &= (1 - 0.10 \times 2)(2 \times 2 + 4 - 0.10 \times 2 \times 4) \\ &= 0.8(4 + 4 - 0.8) = +5.76 \text{ D} \end{aligned}$$

(b) A power of $+5.76$ D corresponds to a focal length

$$f = \frac{1}{P} = \frac{1}{5.76} = 0.173 \text{ m} = +17.3 \text{ cm}$$

(c) The position of the principal point H is determined from Eq. (6o) through the distance

$$H_1H = \frac{0.10}{1 - 0.10 \times 2} = \frac{0.10}{0.80} = 0.125 \text{ m} = +12.5 \text{ cm}$$

It is therefore 12.5 cm to the right of the lens, or 2.5 cm in back of the mirror.

6.7 OTHER THICK MIRRORS

As a second illustration of a thick mirror, consider the thick lens silvered on the back, as shown in Fig. 6H(b). A comparison of this system with the one in diagram (a) shows that Eqs. (6l) to (6p) will apply if the powers P_1 and P_2 are properly defined. For diagram (b), P_1 refers to the power of the first surface alone, and P_2 refers to the power of the second surface as a mirror of radius r_2 in a medium of index n' . In other words,

$$P_1 = \frac{n' - n}{r_1} \quad P_2 = -\frac{2n'}{r_2} \quad \text{and} \quad c = \frac{d}{n'} \quad (6q)$$

With these definitions the power of thick mirror (b) is given by Eq. (6l) and the principal point by Eq. (6o).

The third illustration of a thick mirror consists of a thin lens silvered on the back surface as shown in Fig. 6H(c). This system may be looked upon (1) as a special case of diagram (a), where the mirror has the same radius as the back surface of the thin lens and the spacing d is reduced to zero, or (2) as a special case of diagram (b), where the thickness is reduced to practically zero. In either case Eq. (6l) reduces to

$$\bullet \quad P = 2P_1 + P_2 \quad (6r)$$

and the principal point H coincides with H_1 at the common center of the lens and mirror. P_1 represents the power of the thin lens in air and P_2 the power of the mirror in air, or P_1 represents the power of the first surface of radius r_1 and P_2 represents the power of the second surface as a mirror of radius r_2 in a medium of index n' [see Eq. (6q)].

6.8 SPHERICAL ABERRATION

The discussion of a single spherical mirror in the preceding sections has been confined to paraxial rays. Within this rather narrow limitation, sharp images of objects at any distance may be formed on a screen, since bundles of parallel rays close to the axis and making only small angles with it are brought to a sharp focus in the focal plane. If, however, the light is not confined to the paraxial region, all rays from one object point do not come to a focus at a common point and we have an undesirable effect known as *spherical aberration*. The phenomenon is illustrated in Fig. 6K, where parallel incident rays at increasing distances h cross the axis closer to the mirror. The envelope of all rays forms what is known as a *caustic surface*. If a small screen is placed at the paraxial focal plane F and then moved toward the mirror, a point is reached where the size of the circular image spot is a minimum. This disklike spot is indicated in the diagram and is called the *circle of least confusion*.

The proof that rays from an outer zone of a concave mirror cross the axis inside the paraxial focal point can be simply given by reference to Fig. 6L. According to the law of reflection applied to the ray incident at T , the angle of reflection ϕ'' is equal to the angle of incidence ϕ . This in turn is equal to the angle TCA . Having

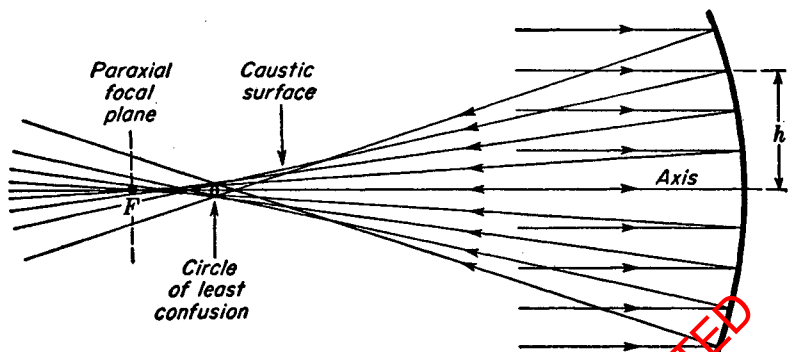


FIGURE 6K
Spherical aberration of a concave spherical mirror.

two equal angles, triangle CTX is isosceles, and hence $CX = XT$. Since a straight line is the shortest path between two points,

$$CT < CX + XT$$

Now CT is the radius of the mirror and equals CA , so that

$$CA < 2CX$$

Therefore

$$\frac{1}{2}CA < CX$$

The geometry of the figure shows that as T is moved toward A , the point X approaches F , and in the limit $CX = XA = FA = \frac{1}{2}CA$.

Over the past years numerous methods of reducing spherical aberration have been devised. If instead of a spherical surface the mirror form is that of a paraboloid of revolution, rays parallel to the axis are all brought to a focus at the same point as in Fig. 6M(a). Another method is the one shown later in Fig. 10Q of inserting a *corrector plate* in front of a spherical mirror, thereby deviating the rays by the proper amount prior to reflection. With the plate located at the center of curvature of the mirror, a very useful optical arrangement known as the *Schmidt system* is obtained. Still a third system, known as a *Mangin mirror*, is shown in Fig. 6M(b). Here a

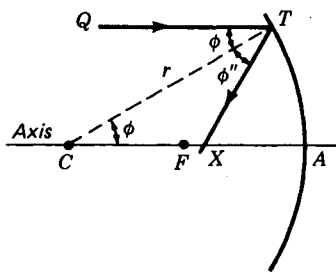


FIGURE 6L
Geometry showing how marginal rays parallel to the axis of a spherical mirror cross the axis inside the focal point.

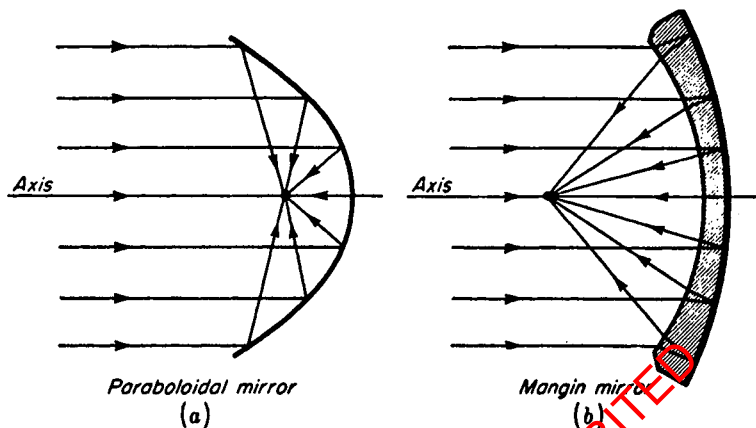


FIGURE 6M

(a) Concave parabolic mirror and (b) concave spherical mirror, corrected for spherical aberration.

meniscus lens is employed in which both surfaces are spherical. When the back surface is silvered to form the concave mirror, all parallel rays are brought to a reasonably good focus.

6.9 ASTIGMATISM

This defect of the image occurs when an object point lies some distance from the axis of a concave or convex mirror. The incident rays, whether parallel or not, make an appreciable angle ϕ with the mirror axis. The result is that, instead of a point image, two mutually perpendicular line images are formed. This effect is known as astigmatism and is illustrated by a perspective diagram in Fig. 6N. Here the incoming rays are parallel while the reflected rays are converging toward two lines S and T . The reflected rays in the vertical or *tangential* plane $RASE$ are seen to cross or to focus at T , while the fan of rays in the horizontal or *sagittal* plane $JAKE$ cross or focus at S . If a screen is placed at E and moved toward the mirror, the image will become a vertical line at S , a circular disk at L , and a horizontal line at T .

If the positions of the T and S images of distant object points are determined for a wide variety of angles, their loci will form a paraboloidal and a plane surface respectively, as shown in Fig. 6O. As the obliquity of the rays decreases and they approach the axis, the line images not only come closer together as they approach the paraxial focal plane but they shorten in length. The amount of astigmatism for any pencil of rays is given by the distance between the T and S surfaces measured along the chief ray.

Equations giving the two astigmatic image positions are*

$$\bullet \quad \frac{1}{s} + \frac{1}{s'_T} = -\frac{2}{r \cos \phi} \quad \frac{1}{s} + \frac{1}{s'_S} = -\frac{2 \cos \phi}{r}$$

* For a derivation of these equations, see G. S. Monk, "Light, Principles and Experiments," Dover Publications, Inc., New York, 1963.

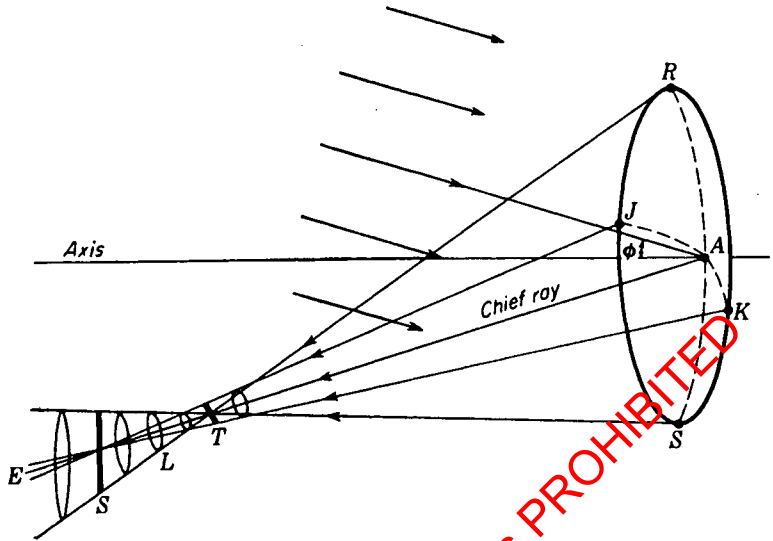


FIGURE 6N

Astigmatic images of an off-axis object point at infinity, as formed by a concave spherical mirror. The lines T and S are perpendicular to each other.

In both equations s and s' are measured along the chief ray. The angle ϕ is the angle of obliquity of the chief ray, and r is the radius of curvature of the mirror.

The Schmidt optical system, which will be discussed later (Fig. 10Q), and the Mangin mirror shown in Fig. 6M(b), constitute instruments in which the astigmatism of a spherical mirror is reduced to a minimum. While the two focal surfaces T and S exist for these devices, they lie very close together and the loci of their mean position (such as L in Fig. 6N) form a nearly spherical surface. The center of this spherical surface is located at the center of curvature of the mirror, as shown in Fig. 10Q.

A paraboloidal mirror is free from spherical aberration even for large apertures but shows unusually large astigmatic $S - T$ differences off the axis. For this reason paraboloidal reflectors are limited in their use to devices that require a small angular spread, such as astronomical telescopes and searchlights.

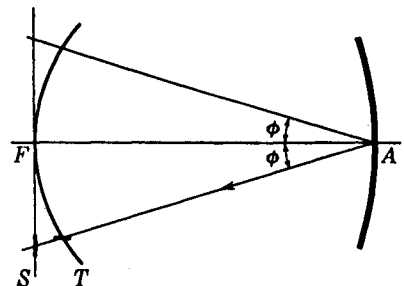


FIGURE 6O

Astigmatic surfaces for a concave spherical mirror.